

Weekly Report

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Research Topic: Inferring Hearthstone Player Play-Style From Game Logs

1. Recap of last week

Last week I simulated many games and saved each one as a log file. From each log I computed simple behavioral statistics (for example, damage to the enemy hero per turn). The output is a table where one row is one game and the columns are the extracted metrics. I then ran both unsupervised (KMeans) and supervised (LDA) learning on this table.

Within a single deck the results were strong — the five play-styles (aggro, control, fatigue, midrange, ramp) separated clearly once I pooled enough games (ARI around 0.4–0.6 at N=10, and near 0.99 at N=120). But cross-deck (cross-domain) was weak: when I trained on one deck and tested on another, the score dropped close to chance. Only the fatigue style kept transferring; the others fell apart. I also tried deck-normalization (measuring each game as its distance from that deck's average player), which roughly doubled the cross-deck score (about 0.08 → 0.23 at N=10) but still did not fix it.

2. This week, Part 1 — Testing the "card-sequence-only" idea

You suggested a good first check: simulate labeled agents, but keep only the information the real human dataset has — the card-play sequence, with no board, health, hand, or combat data. The logic is a feasibility gate:

- If labeled simulated agents cannot be told apart using card-sequence-only features, then the real human dataset is impossible, and there is no point continuing.
- If they can be told apart, then the human dataset is worth trying.

So I rebuilt the feature table using only the metrics that can be computed from card-play order alone. This is a small subset of my earlier metrics:

- Still computable (order/cost based): average card cost, minion fraction, cards played per turn, turn of the first minion, and similar.
- Lost (need board/health/hand/combat): face damage per turn, face-attack ratio, attacks per turn, enemy minions killed, board size, damage taken, hero-power tempo, and so on.

Result: classification on this card-sequence-only dataset was poor. The reason is simple — the usable feature set is much smaller than before, so most of the signal that separated the styles is

gone. I have not yet found new metrics that can rebuild that lost signal from card order alone. This is genuinely hard, and I do not have a good idea for it yet.

3. This week, Part 2 — Deciding to fix cross-domain first

Because of the above, I think the better next move is to first make cross-domain transfer work on the richer simulated data, before pushing harder on the feature-starved card-sequence data. My reasoning:

- If cross-domain works on the rich data, that is direct evidence the model is capturing the player's play-style, not just the deck. This is exactly RQ3 — "is it the player, or the deck?"
- The card-sequence-only case is even harder than the full-log case (fewer features). If I cannot solve cross-domain with rich features, solving it with poor features is very unlikely. So it makes sense to solve the easier version first.

4. This week, Part 3 — Expanding the simulation to all 9 decks

In the last report, Agent 1 played only 2 of the 9 decks the simulator provides (RenoKazakusMage and AggroPirateWarrior). So my cross-deck test only had one deck pair. This week I am re-simulating so Agent 1 plays all 9 decks. The reason: cross-domain evidence is much stronger with more domains. With 2 decks I get only one transfer direction, which could be a fluke of those specific decks. With 9 decks I get many train-deck / test-deck combinations, so the cross-domain result becomes more trustworthy and more meaningful.

5. Plan for next week

- Focus on improving cross-domain transfer using the statistical metrics on the new 9-deck data.
- Diagnose why the metrics fail to transfer — for example, check whether some metrics point in opposite directions on different decks, keep only the metrics that order the styles consistently across decks, and keep improving deck-normalization.
- After cross-domain is better, return to the card-sequence-only dataset with its limited features.

6. Interesting paper I found

I found this paper fascinating and very close to my project: "*Detecting Individual Decision-Making Style: Exploring Behavioral Stylometry in Chess.*" [\[1\]](#) Why it is exciting for my work:

- Same core problem, different game. They identify a person from their decisions alone — this is exactly "infer play-style from game logs," just in chess instead of Hearthstone.
- Decisions-only input, like my human case. They work only from the moves a player made, with no extra engine state. This is the same limitation I face with the human dataset (card sequence only), and the paper shows the problem is solvable from decisions alone.
- Pooling many games, like my RQ2. They identify a player from among thousands of candidates with 98% accuracy using only about 100 games, not a single game. This matches my finding that one game is too noisy and the style only appears after averaging many games.
- Cross-distribution transfer, like my cross-deck problem. They train on amateur play and still generalize to Grandmaster players, despite the big differences between them. This is a domain-shift success — the same kind of transfer I am struggling with across decks. It suggests style can survive a domain shift if the representation is good, which supports my plan to fix the feature representation instead of giving up on cross-domain.

One difference: their method uses a large neural network (a transformer) to learn its own move embeddings, while I use simple, human-readable statistics. So for now the paper is proof that the goal is reachable, and a possible direction for much later — this week I am keeping the work on the statistical metrics.

Reference

[1] R. McIlroy-Young, R. Wang, S. Sen, J. Kleinberg, and A. Anderson, "Detecting Individual Decision-Making Style: Exploring Behavioral Stylometry in Chess," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [Online]. Available: [arXiv:2208.01366](https://arxiv.org/abs/2208.01366).